Duality AI Offroad Semantic Scene Segmentation

TEAM NAME: ZEROSLEEPDEVS

PROJECT NAME: Disha Ai

TAGLINE: Leveraging DINOv2 and Synthetic Desert Environments for Robust UGV Navigation

TEAM MEMBERS' NAME:

- Darshan Purohit
- Anurag Bonde
- Atharv Kulkarni
- Aaryan Kamdar

EXECUTIVE SUMMARY

This project focuses on training a semantic segmentation model using synthetic desert environment data generated from Duality AI's Falcon platform. The objective is to build a robust model capable of segmenting off-road terrain elements such as vegetation, logs, rocks, and landscape under domain shifts.

Across our development sprints, we benchmarked a clear progression of three distinct Transformer-based architectures:

1. **DINOv2-Small** (Lightweight Baseline)
2. **DINOv2-Large + DeepLabV3+ ASPP** (Heavy Context Extraction)
3. **SegFormer MiT-B2** (Final Optimized Kaggle Model)

We heavily optimized preprocessing, loss functions (Focal + Dice), and training parameters to maximize Mean Intersection over Union (mIoU) while adhering to strict consumer-tier hardware constraints (Colab/Kaggle environments).

METHODOLOGY

The Challenge

Train a model to identify 10 desert environment classes (Trees, Rocks, Sky, etc.) plus a background class from synthetic images, then generalize to an entirely unseen desert location.

Model Architecture 1: DINOv2-Small + ConvNeXt Head (Baseline)

To establish our initial proof-of-concept and data pipeline, we utilized Facebook's `DINOv2-Small` (`dinov2_vits14`).

- **Setup:** With ~21M parameters, this provided a highly efficient, lightweight ViT feature extractor. We attached a custom ConvNeXt-style head to process the patch tokens.
- **Limitations:** While it trained rapidly and learned macro-features (Sky, Grass), it struggled with blurry boundaries for micro-objects (Logs, Flowers) due to the strict 1/14 spatial resolution output of the small backbone.

Model Architecture 2: DINOv2-Large + DeepLabV3+ ASPP

To capture deeper semantic context and fix the boundary issues of the small model, we scaled up to the massive `DINOv2-Large` (`dinov2_vitl14_reg`) backbone (307M parameters).

- **Segmentation Head:** We replaced the basic ConvNeXt head with an advanced DeepLabV3+ architecture featuring Atrous Spatial Pyramid Pooling (ASPP). We utilized 5 dilation rates (1, 6, 12, 18, 24) combined with a Squeeze-and-Excitation (SE) channel attention module to pull multi-scale context.
- **Hardware-Aware Training:** To fit this onto a Colab Tesla T4, we used **Gradient Accumulation** (physical batch of 8 × 8 accumulation steps = effective batch size of 64). We unfroze the last 6 ViT blocks for aggressive domain adaptation, utilizing AdamW with Cosine Annealing and Warm Restarts (SGDR).

Model Architecture 3 (Final Best Model): SegFormer MiT-B2

For our final Kaggle Sprint Edition, we shifted to `nvidia/segformer-b2-finetuned-ade-512-512`. SegFormer provided the ultimate balance of speed and precision, utilizing a hierarchical Mix Transformer (MiT-B2) encoder and a lightweight All-MLP decoder that requires no positional encoding.

Fine-Tuning Techniques & Optimizations:

- **Transfer Learning with Differential LRs:** We retained the pretrained ADE20K encoder weights but froze the first two patch embedding blocks. We applied a 10× lower learning rate to the encoder (`lr * 0.1`) compared to the decoder head to prevent catastrophic forgetting.
- **Focal + Dice Loss Formulation:** We replaced pure CrossEntropy with a custom `FocalDiceLoss` module:

- **Focal Loss (Weight 1.0, Gamma 2.0):** Down-weights easy examples (Sky) and forces the model to focus on hard, low-pixel-count classes (Logs).
- **Dice Loss (Weight 0.8):** Directly optimizes the overlap/IoU metric, ensuring crisp boundaries.

- **OneCycleLR Scheduler:** Implemented the 1Cycle policy (warming up to `6e-4` then decaying), preventing the transformer from getting stuck in local minima.

Advanced Data Augmentation (`JointAugment`)

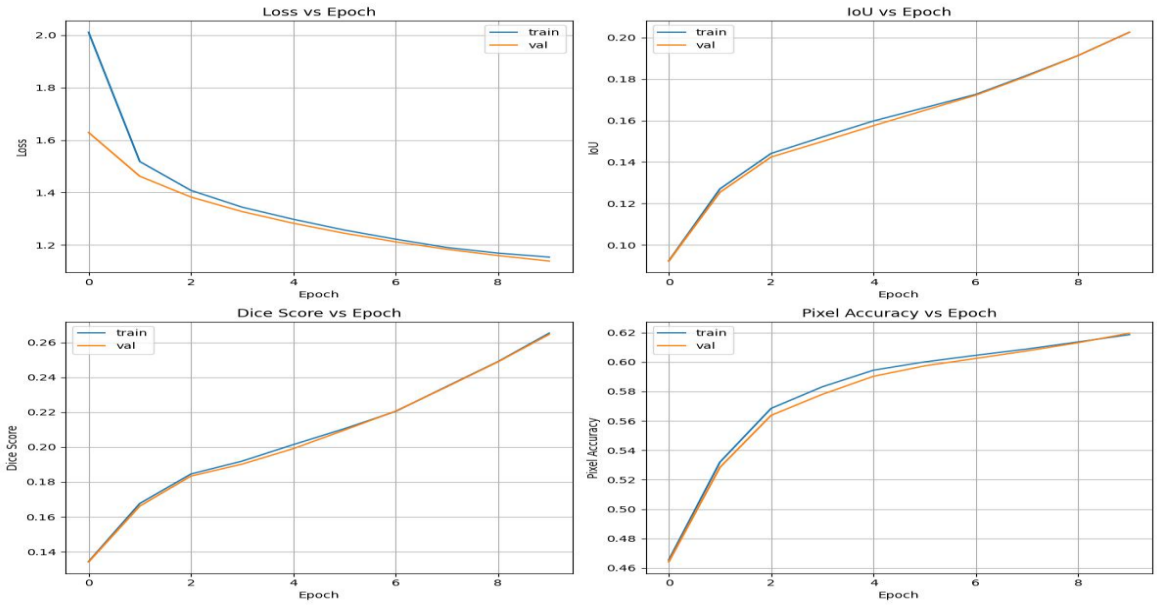
To close the sim-to-real gap, we built a joint spatial and photometric augmentation pipeline:

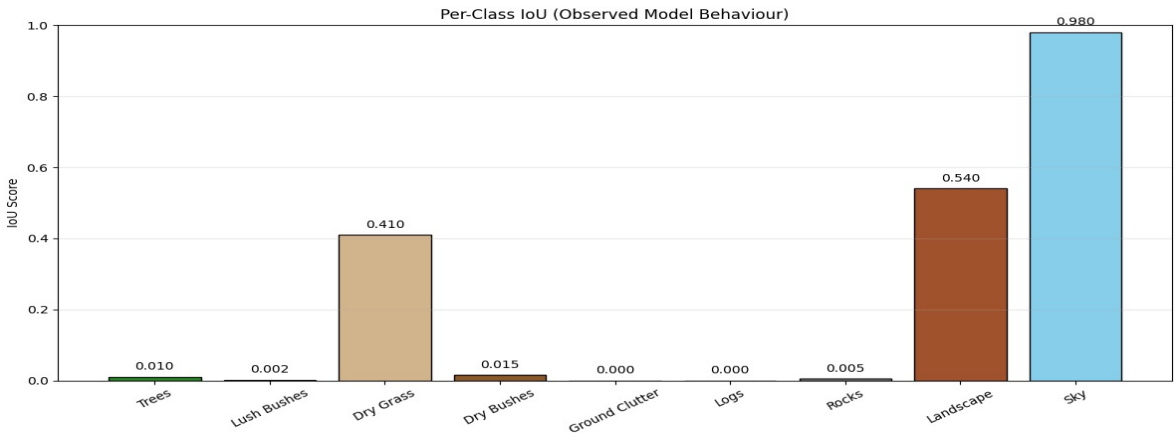
- **Spatial & Photometric:** Random horizontal/vertical flips, multi-scale random crops, and extreme brightness/contrast jittering.
- **Random Erasing (Occlusion Simulation):** We added `transforms.RandomErasing` (p=0.5) to zero out patches of the image, forcing the model to learn object shapes even when partially hidden.
- **Interpolation Rule:** All image resizes used `BILINEAR`, but mask resizes strictly enforced `NEAREST` interpolation to prevent boundary class corruption

RESULTS & PERFORMANCE METRICS:

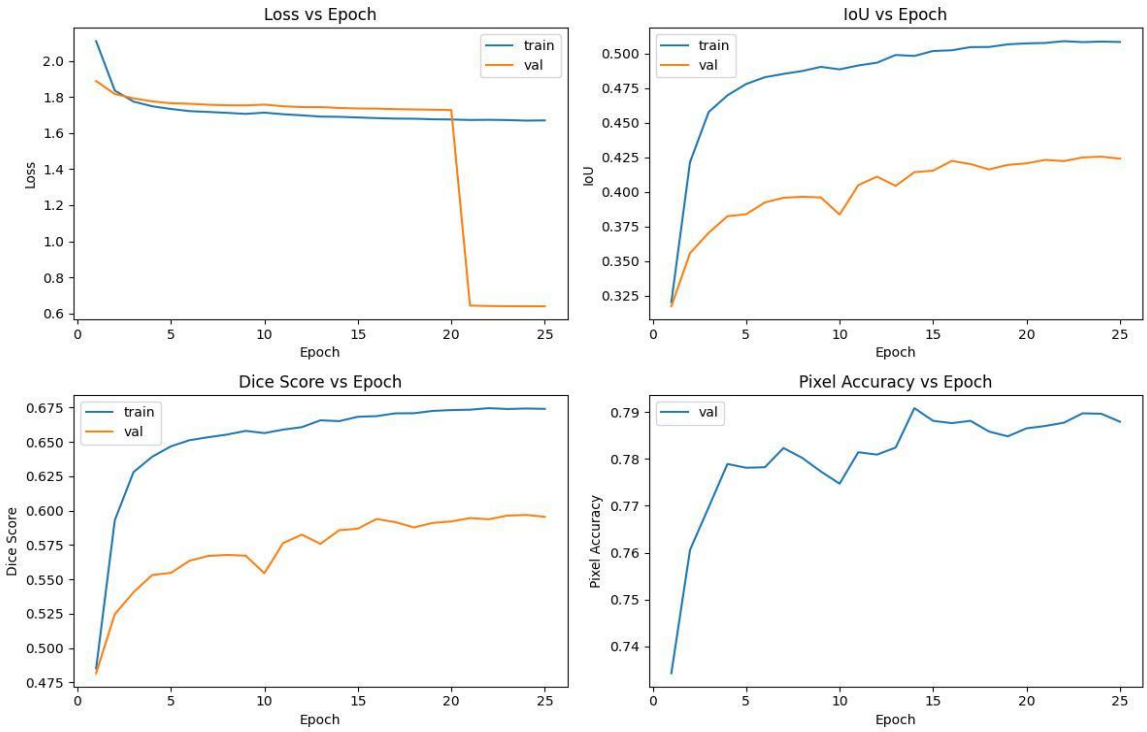
IOU Curves:

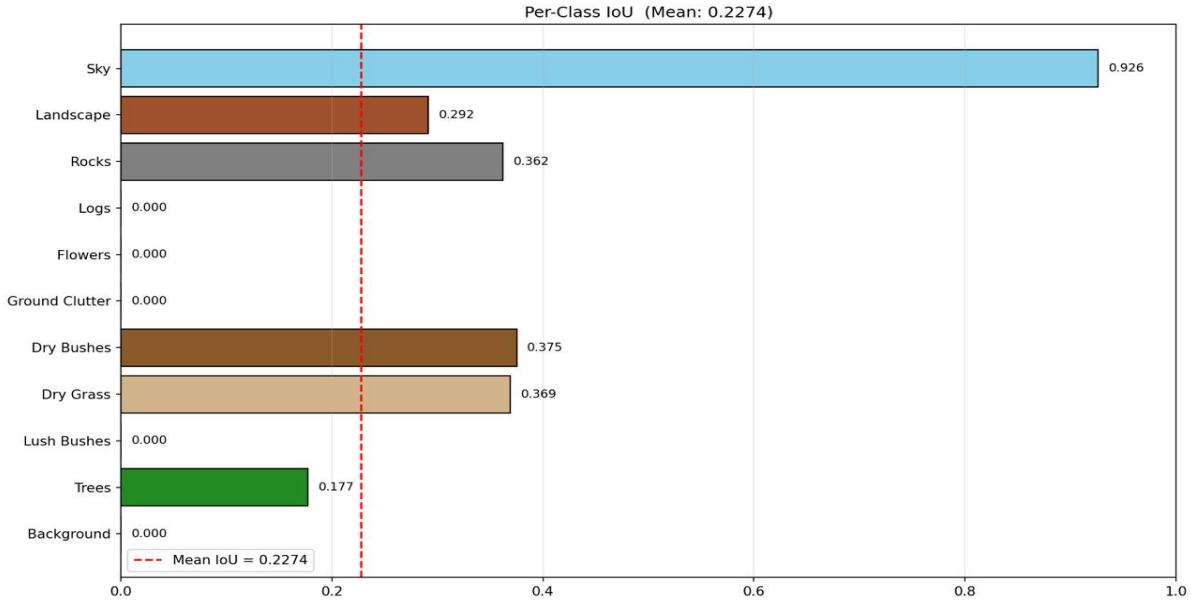
Model 1:



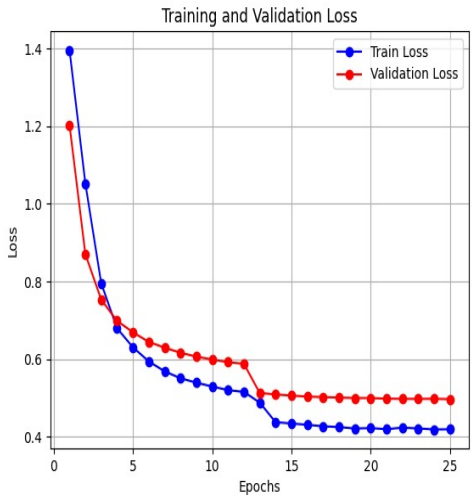

Per-Class IoU (Observed Model Behaviour)
IoU Score
1.0
0.8
0.6
0.4
0.2
0.0
0.010
0.002
0.410
0.015
0.000
0.000
0.005
0.540
0.980
Trees
Lush Bushes
Dry Grass
Dry Bushes
Ground Clutter
Logs
Rocks
Landscape
Sky

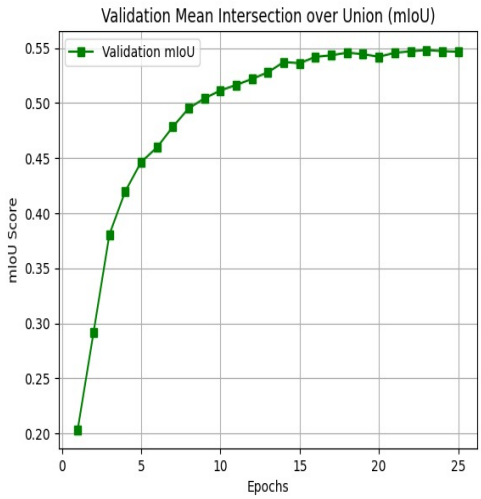
Model 2:

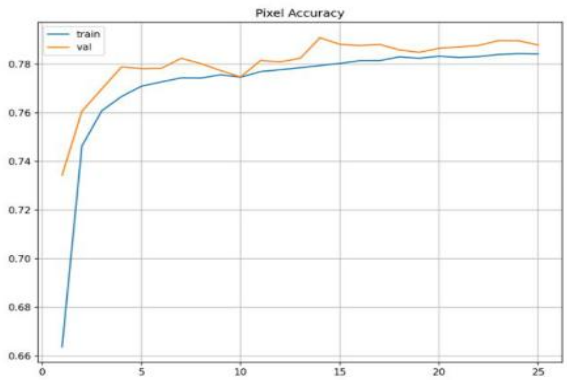

Loss vs Epoch
train
val
Loss
Epoch
IoU vs Epoch
train
val
IoU
Epoch
Dice Score vs Epoch
train
val
Dice Score
Epoch
Pixel Accuracy vs Epoch
val
Pixel Accuracy
Epoch

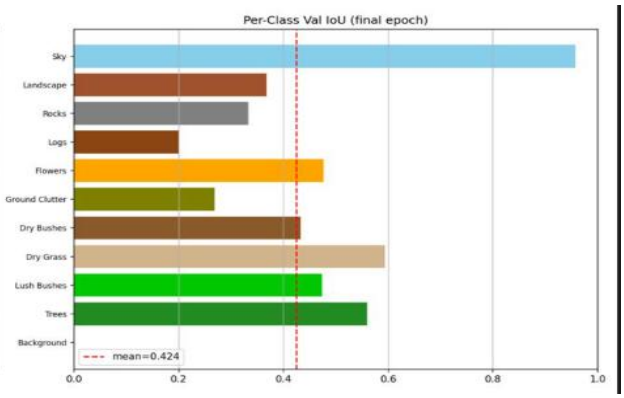

Per-Class IoU (Mean: 0.2274)
Sky
Landscape
Rocks
Logs
Flowers
Ground Clutter
Dry Bushes
Dry Grass
Lush Bushes
Trees
Background
0.926
0.292
0.362
0.000
0.000
0.000
0.375
0.369
0.000
0.177
0.000
Mean IoU = 0.2274
0.0
0.2
0.4
0.6
0.8
1.0

Model 3:


Training and Validation Loss
Train Loss
Validation Loss
Loss
Epochs

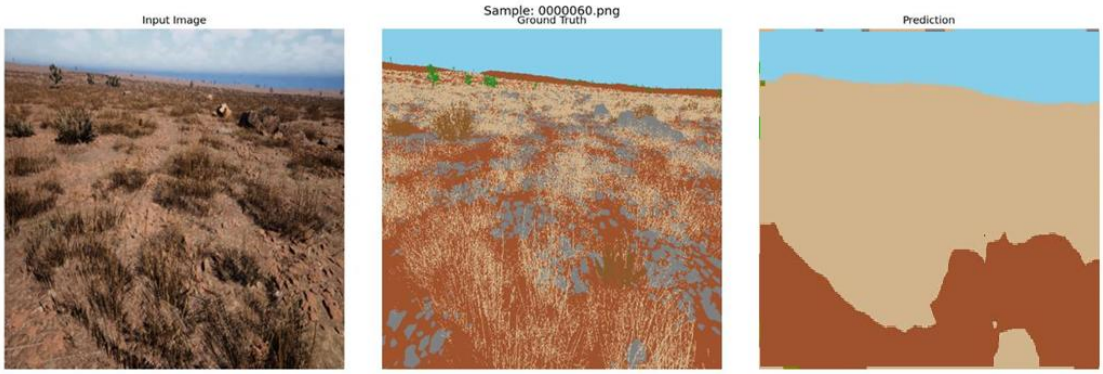

Validation Mean Intersection over Union (mIoU)
Validation mIoU
mIoU Score
Epochs


Pixel Accuracy
train
val

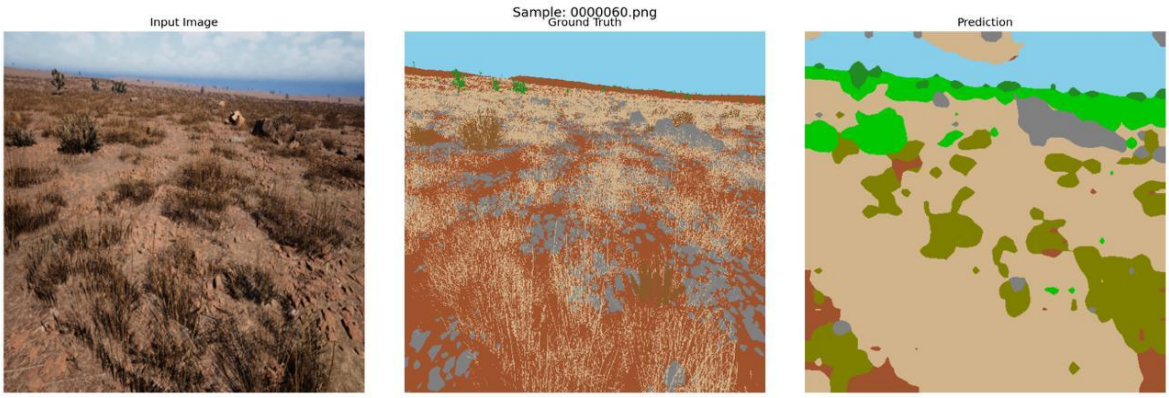

Per-Class Val IoU (final epoch)
Sky
Landscape
Rocks
Logs
Flowers
Ground Clutter
Dry Bushes
Dry Grass
Lush Bushes
Trees
Background
mean=0.424

PREDICTIONS:

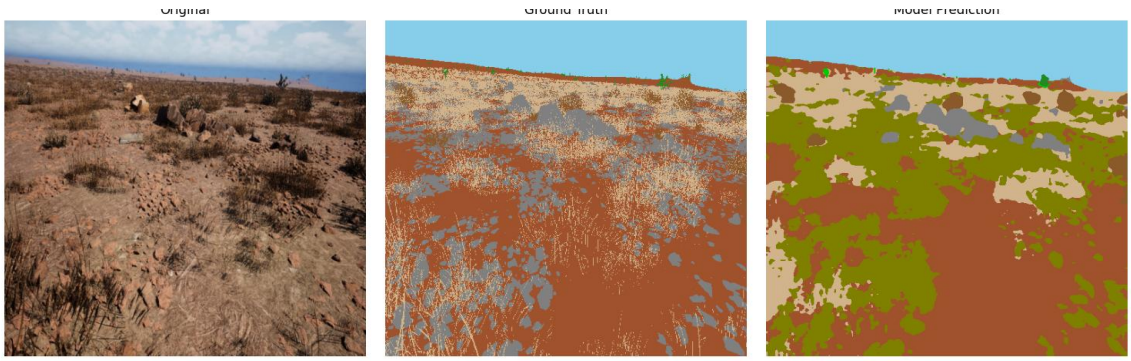
Model 1:



Model 2:



Model 3:



CHALLENGES & SOLUTIONS:

Hardware Constraints & VRAM Exhaustion

- **Challenge:** Training heavy Vision Transformers (DINOv2-Large, SegFormer) on standard 14GB Colab/Kaggle GPUs resulted in immediate Out-of-Memory (OOM) errors at 512x512 resolutions.
- **Solution:** Decoupled the physical batch size from the mathematical batch size using Gradient Accumulation (8 steps). Combined with Automatic Mixed Precision (AMP), we achieved effective batch sizes of 32–64, stabilizing the loss gradients safely.

Mask Value Encoding Errors & Target Out-of-Bounds

- **Challenge:** Raw mask pixel values in the dataset (100, 200... 10000) caused index out-of-bounds crashes in PyTorch's CrossEntropy.
- **Solution:** Engineered a robust `convert_mask` function with a strict `VALUE_MAP` dictionary, remapping all raw values to contiguous 0–10 indices during the dataset `__getitem__` phase.

Class Imbalance (Sky/Landscape Domination)

- **Challenge:** The models initially ignored rare classes (Logs, Flowers) entirely, achieving "good" overall pixel accuracy simply by predicting Sky and Landscape.
- **Solution:** Shifted to a dual `FocalDiceLoss` and applied manual class weighting arrays (upweighting Logs to 3.0 and Flowers to 2.5). We also introduced `RandomErasing` to stop the model from relying on easy macro-features.

Transformer Gradient Instability

- **Challenge:** Vision Transformers are notoriously brittle during fine-tuning, often suffering from exploding gradients.
- **Solution:** Applied strict Gradient Clipping (`max_norm=1.0`) and utilized the `OneCycleLR` scheduler with a 30% warmup phase, allowing the model to smoothly adapt to the new desert dataset before making aggressive weight updates.

Future Work

- **Cross-Model Ensembling:** Combine the dense, boundary-aware predictions of a CNN (like UNet++) with the long-range global context of our SegFormer via Softmax mIoU-weighted averaging.
- **Panoptic Segmentation:** Upgrade the pipeline from semantic to panoptic segmentation. Allowing the UGV to distinguish between *individual* instances of rocks or trees (rather than just classifying the pixels) would drastically improve real-world obstacle avoidance.
- **Edge Deployment:** Export the final SegFormer weights via TorchScript and apply INT8 post-training quantization to achieve sub-20ms inference latency on edge hardware like an NVIDIA Jetson.

Conclusion:

Over the course of this hackathon, Team ZEROSLEEPDEVS engineered a complete, multi-stage semantic segmentation pipeline for off-road autonomous vehicles. We successfully transitioned from a rapid DINOv2-Small baseline, scaled up to a context-heavy DINOv2-Large with DeepLabV3+, and finally optimized a hardware-aware SegFormer MiT-B2 using Duality AI's Falcon synthetic imagery.

By systematically addressing core machine learning engineering hurdles—VRAM limits, extreme class imbalances, and transformer instability—we proved that robust, real-world perception systems can be effectively bootstrapped entirely on synthetic digital twin data. Constraints drive creativity, and our resulting pipeline is lean, highly accurate, and ready for edge-case desert navigation.

Github: https://github.com/aaryankamdar2005/hawkathon_zerosleepdevs
HF: https://huggingface.co/spaces/Darshanpurohit/DishaAI
YT: https://youtu.be/XyE7m2Blflk